

- CSC301: DATA STRUCTURES AND ALGORITHMS
- Complete Lab Programs - Labs I to III

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■      Section   : B

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- All 33 Programs | Theory | Algorithm | Output
- CSC301 DSA Lab | Tribhuvan University

0 1 4

```
5 6 0
Product A*I = A:
1 2 3
0 1 4
5 6 0
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sachin@MacBook-Air DSA_codes % gcc labno8.c -o labno8 && ./labno8
Enter n: 3
2 1 -1 8
-3 -1 2 -11
-2 1 2 -3
x[1] = 2.0000
x[2] = 3.0000
x[3] = -1.0000
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>>> LAB II – Stack & Queue

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sachin@MacBook-Air DSA_codes % gcc labno9.c -o labno9 && ./labno9
1 (Push) -> 10
1 (Push) -> 20
1 (Push) -> 30
2 (Pop)
3 (Peek)
4 (Display)
Pushed: 10
Pushed: 20
Pushed: 30
Popped: 30
Top element: 20
Stack: 20 10
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sachin@MacBook-Air DSA_codes % gcc labno10.c -o labno10 && ./labno10
1 (Push) -> 5
1 (Push) -> 15
1 (Push) -> 25
2 (Pop)
3 (Peek)
Pushed: 5
Pushed: 15
Pushed: 25
Popped: 25
Top element: 15
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sachin@MacBook-Air DSA_codes % gcc labno11.c -o labno11 && ./labno11
Enter infix expression: A+B*C-D
Postfix expression: ABC*+D-
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sachin@MacBook-Air DSA_codes % gcc labno12.c -o labno12 && ./labno12
Enter postfix expression: 23*54-+
Result = 7.0000
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sachin@MacBook-Air DSA_codes % gcc labno13.c -o labno13 && ./labno13
1 (Enqueue) -> 100
1 (Enqueue) -> 200
1 (Enqueue) -> 300
2 (Dequeue)
3 (Display)
Enqueued: 100
Enqueued: 200
Enqueued: 300
Dequeued: 100
Queue: 200 300
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```

```
sachin@MacBook-Air DSA_codes % gcc labno14.c -o labno14 && ./labno14
Enqueue: 10 20 30 40 50
Enqueue: 60 (Full)
Dequeue
Enqueue: 60 (reuse slot)
Enqueued: 10 20 30 40 50
Queue Full!
Dequeued: 10
Enqueued: 60
Queue: 20 30 40 50 60
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```

```
sachin@MacBook-Air DSA_codes % gcc labno15.c -o labno15 && ./labno15
Enqueue 5 (pri:3)
Enqueue 15 (pri:1)
Enqueue 25 (pri:2)
Dequeue
Enqueued: 5 Priority: 3
Enqueued: 15 Priority: 1
Enqueued: 25 Priority: 2
Dequeued (highest priority): 5
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```

>>> LAB III – Linked Lists, Recursion & Sorting

```
sachin@MacBook-Air DSA_codes % gcc labno16.c -o labno16 && ./labno16
Insert Beg: 10
Insert Beg: 20
Insert End: 30
Insert Pos 2: 25
Display
Delete: 20
Display
List: 20 -> 10 -> NULL
List: 20 -> 10 -> 30 -> NULL
List: 20 -> 25 -> 10 -> 30 -> NULL
Deleted: 20
List: 25 -> 10 -> 30 -> NULL
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```

```
sachin@MacBook-Air DSA_codes % gcc labno17.c -o labno17 && ./labno17
Push: 5
Push: 10
Push: 15
Pop
Peek
Display
Pushed: 5
Pushed: 10
Pushed: 15
Popped: 15
Top: 10
Stack: 10 -> 5 -> NULL
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```

```
sachin@MacBook-Air DSA_codes % gcc labno18.c -o labno18 && ./labno18
Enqueue: 1
Enqueue: 2
Enqueue: 3
Dequeue
Display
Enqueued: 1
Enqueued: 2
Enqueued: 3
Dequeued: 1
Queue: 2 -> 3 -> NULL
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```

```
sachin@MacBook-Air DSA_codes % gcc labno19.c -o labno19 && ./labno19
```

```
Enter n: 5
Factorial(5) = 120
Fibonacci(7) = 13
GCD(12, 8) = 4
Tower of Hanoi (3 disks):
  A -> C
  A -> B
  C -> B
  A -> C
  B -> A
  B -> C
  A -> C

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sachin@MacBook-Air DSA_codes % gcc labno20.c -o labno20 && ./labno20
Enter array size: 6
Elements: 64 34 25 12 22 11
Bubble Sort: 11 12 22 25 34 64
Selection Sort: 11 12 22 25 34 64
Insertion Sort: 11 12 22 25 34 64

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sachin@MacBook-Air DSA_codes % gcc labno21.c -o labno21 && ./labno21
Enter array size: 6
Elements: 38 27 43 3 9 82
Merge Sort: 3 9 27 38 43 82
Quick Sort: 3 9 27 38 43 82

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sachin@MacBook-Air DSA_codes % gcc labno22.c -o labno22 && ./labno22
Sorted array: 1 3 5 7 9
Enter search key: 7
Sequential Search: Found 7 at index 3
Binary Search: Found 7 at index 3

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sachin@MacBook-Air DSA_codes % gcc labno23.c -o labno23 && ./labno23
Enter array: 10 20 5 15 25 30
Hash Table (size 7):
[0] -> -
[1] -> 15
[2] -> 30
[3] -> 10
[4] -> 25
[5] -> 5
[6] -> 20
Hashing complete.

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